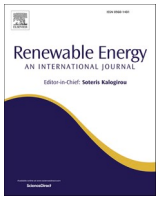
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A robust MPPT method based on optimizable Gaussian process regression and high order sliding mode control for solar systems under partial shading conditions

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ABSTRACT

This study proposes a new hybrid approach for maximum power point tracking (MPPT) under partial shading conditions. The approach consists of a high order sliding mode controller (HOSMC) and a hyperparameters optimized Gaussian Process Regression (GPR). The proposed hybrid method consists of two stages. In the first stage, the model created from real-time data was trained using optimized hyperparameters GPR method, one of the machine learning methods. Moreover, at this phase, a model is constructed utilizing temperature and irradiance values as inputs, with the output being the voltage value. Using this model, the reference voltage values required for HOSMC in different scenarios were generated. In the second stage, the error expression was obtained by comparing the generated reference voltage value with the PV input voltage value. The control signal is generated to minimize the error value and MPPT is performed. The Incremental Conductance (INC) algorithm, a traditional optimization technique, and the Gray Wolf Optimization (GWO), a metaheuristic optimization algorithm, were used to compare the suggested hybrid method in order to show its efficacy. Data from different months and seasons (February, May, August and October) were used to determine the scenarios. The irradiance and temperature values used in the scenarios were obtained from real-time data. Simulation studies were carried out using Matlab/Simulink. When the proposed hybrid method was compared with the INC and GWO algorithms, it was seen that it had the fastest convergence time and the largest efficiency value in all scenarios.

1. Introduction

Due to the harmful effects of fossil energy fuels on the environment, the use of renewable energy sources is increasing day by day. Solar energy, which is an inexhaustible energy source among renewable energy sources, has become the center of attention due to its advantages such as unlimited potential and low maintenance [1]. In addition, solar power generation does not cause environmental pollution or noise and requires less maintenance. Solar energy systems consist of one or more photovoltaic (PV) panels. The most important disadvantage of energy systems consisting of these panels is that they are sensitive to environmental factors. Depending on environmental factors, there may be changes in the amount of energy produced. Maximum power point tracking (MPPT) algorithms have been developed to get rid of this drawback. Using these algorithms, MPPT can be performed even in PV panels operating under different environmental conditions.

PV panels can be connected in parallel to generate higher current values or in series to generate higher voltages. A PV panel has a single maximum power point (MPP) that is determined by the temperature and level of irradiance. The temperature and irradiance values of individual PV panels in systems with multiple panels may differ based on environmental factors. This situation is known as partial shading. There are several local maximum power points (LMPP) and one global maximum power point (GMPP) in the power-voltage (P-V) characteristics of such systems. Power generation plants use a number of PV panels, so it's critical that the system function at GMPP.

In the literature, the most widely used classical methods for PV systems to operate at GMPP are the perturb&observe (P&O) [2] and the incremental conductance (INC) [3] algorithms. Although these algorithms exhibit high performance at constant irradiance and temperature, they cannot reach the desired GMPP point under partial shading conditions. Metaheuristic approaches such as harris hawk optimization (HHO) [4] and gray wolf algorithm (GWO) [5] have been used to overcome this shortcoming of classical methods. However, the most

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important shortcoming of these methods is that they involve complex calculations. Artificial intelligence-controller based approaches such as artificial neural networks (ANN), fuzzy logic controller (FLC), deep learning (DL) and machine learning (ML) methods are also used for GMPP monitoring [6–9].

In order to eliminate the disadvantages of classical and metaheuristic optimization algorithms, these methods have been modified or used as hybrid [10–12]. The main purpose of hybrid methods is to combine the superior features of different algorithms to obtain more successful results. In Ref. [13], a hybrid method was suggested by combining GWO with arithmetic optimization algorithm (AOA). The performance of this hybrid method is compared with AOA, GWO, particle swarm optimization (PSO) and P&O under different environmental conditions. According to the simulation studies and real-time results obtained, they stated that the proposed method has higher performance. Baraean et al. proposed a hybrid method for partial shading conditions and variable load case. In this hybrid method, they designed an MPPT controller integrating neural network (NN) and adaptive terminal sliding mode controller (ATSMC). They reported that this controller structure reduces the chattering effect under different disturbances and load conditions [14]. Omar et al. suggested an approach consisting of fractional order incremental conductance (FOINC) and fractional order proportional integral (FOPI) controller to provide high tracking accuracy at different irradiance levels [15]. Divyasharon et al. proposed the logarithmic seagull artificial neural network-based improved perturb and observe maximum power point tracking (LSANN-IPOMPPT) algorithm to track MPP under different atmospheric conditions. In this hybrid approach, they utilized the LSANN method to calculate the desired current and voltage values. They used the IPOMPPT algorithm for maximum power transfer to the load. They stated that this method, applied to the different converter structure they used in the design, has higher performance than other algorithms [16]. Although the irradiance and temperature values generally used in the literature are user defined values, the actual data are different from these values. For this reason, it is very important to use real data obtained through sensors with high accuracy rates in the studies. In their study, Bayarassou et al. stated that the data obtained from sensors with low accuracy will not be accurate. For this purpose, they used fuzzy linear regression and fuzzy least squares approaches to determine imprecise model parameters. They reported that both techniques gave similar results and MPP was successfully monitored [17].

Yılmaz et al. proposed a new hybrid MPPT algorithm for PV systems operating under both equal irradiance and partial shading conditions. They used a two-stage structure while designed the hybrid method. In the first stage, the voltage value that will enable the PV system to operate in GMPP was estimated using machine learning methods. In the second stage, the optimal duty cycle was tried to be adjusted with the super twisting sliding mode controller (STSMC). In their study, they considered statically changing irradiance and temperature conditions in three scenarios and dynamically changing environmental conditions in the fourth scenario. The proposed hybrid method was compared with the P&O algorithm and the cuckoo search algorithm (CSA). As a result of the simulation studies, the superiority of the proposed hybrid method was confirmed [9].

Yılmaz et al. proposed a new hybrid method with high efficiency for PV systems. In order to estimate the voltage value in GMPP, Linear Regression, Support Vector Machine and optimized gaussian process regression (GPR) methods were compared. In the proposed hybrid method, optimized GPR was used for estimating the reference voltage value and FLC was used to adjust the converter duty cycle. In the study using real-time data from Bayburt province, the proposed hybrid method was compared with PSO and flying squirrel search optimization (FSSO) algorithms. The proposed hybrid method was evaluated for four different scenarios where GMPPs were in different regions in P-V characteristics and a scenario with dynamic irradiance-temperature changes [18].

Khan et al. proposed a new hybrid method for sudden changes in irradiance, temperature and load conditions in PV systems. They estimated the voltage value in MPP using GPR. They used backstepping super-twisting sliding mode control (BSTSMC) as the controller. The proposed hybrid method was compared with P&O algorithm, PID controller and classical SMC. According to the results obtained from the simulation studies, it was observed that the proposed hybrid method gave more successful results compared to other methods in terms of performance criteria such as fast dynamic response, convergence in finite time, minimum oscillation and efficiency [19].

Huang et al. proposed a hybrid method for MPPT under partial shading conditions. They combined the GPR method and Jaya algorithm to create the hybrid method. They compared the proposed hybrid method with PSO and Jaya algorithms. In their simulation studies for different PV panel configurations, it was observed that the proposed hybrid method has higher performance in terms of convergence time and dynamic efficiency [20].

Kchaou et al. proposed a second order SMC based on super twisting algorithm for PV systems. They compared the proposed method with the classical SMC and RUCA algorithm for three different scenarios. In the first scenario, standard test conditions were considered, in the second scenario, the dynamic change of the irradiance value was considered, and in the third scenario, real-time data were considered. They verified the effectiveness of the proposed method with simulation studies [21].

A third-order sliding mode controller (TOSMC) technique for MPPT under variable environmental and climatic conditions is presented in Ref. [22]. In the study, the panel temperature value was determined as a fixed $25^{\circ}C$. The irradiance values were used between user-defined 600 W/m^2. and 1000 W/m^2. A comparison is made between the suggested approach and SMC concerning performance metrics like efficiency and steady-state error. They claimed that by utilizing TOSMC, they were able to lessen the chattering effect that is present in the classical SMC controller. The simulation model employed a single panel with variable irradiance values. Also, they mentioned that TOSMC improves power quality in various environmental conditions. They also observed from the results that TOSMC has less oscillation in MPP and performs MPPT with higher efficiency compared to classical SMC.

In this study, optimizable GPR is combined with TOSMC, unlike the aforementioned study, more than one panel was used and partial shading was analyzed using real-time data. In most of the past studies, irradiance and temperature values were realized using user-defined values such as 1000 W/m^2-25 °C, 600 W/m^2 −20 °C. However, the actual data varies depending on time of day. It is very important to use real-time data to obtain the closest results to the reality in PV systems. Real-time hourly temperature and irradiance data of Aziziye district in Erzurum province, where solar power plants are located intensively, are used for the reasons mentioned above. In order to demonstrate the effectiveness of the method, one month from each season of 2022 (February, May, August and October) is selected and real-time data for these months are used. Simulation studies were carried out using these data in six different scenarios. In the simulation studies, a highly efficient hybrid method was used for different scenarios by combining GPR with optimized hyperparameters in machine learning algorithms and a TOSMC. The most important advantages of this hybrid method are that the MPPT speed is high, it does not involve complex mathematical calculations, the chattering problem is minimized and the trained panel performs MPPT in different environmental conditions. Using real-time data from the Erzurum/Aziziye region, the suggested method is compared with the GWO and the INC algorithms in terms of performance metrics like efficiency and convergence time.

One of the most important differences between the study in Ref. [9] and this study is due to the controllers. While there is a second-order sliding mode in Ref. [9], a third-order sliding mode is used in this study. This provides higher controller performance in complex systems that generate electricity from PV panels. In addition, the TOSMC reduces

the effect of discrete signals on the system and provides a smoother control signal. In Ref. [9], training was performed using data between April 1, 2021 and March 31, 2022. In this study, the training phase was carried out with four months of data from February, May, August and October of 2022. In Ref. [9], irradiance, temperature and voltage values were used directly in the training process. In this study, irradiance, temperature and voltage values were used by standardizing them. In this study, unlike the study in Ref. [9], the situation of simultaneous dynamic changes in irradiance, temperature and load values was tested and the performance of the proposed hybrid method was verified.

In [22], TOSMC structure was used to perform MPPT. The performance of this controller was tested for a system consisting of a single PV panel in the case of constant temperature and irradiance changes at certain intervals. Unlike their work, the GPR-TOSMC method was proposed in this study. The reference voltage value required for this proposed method was obtained using GPR with optimized hyperparameters from machine learning methods. TOSMC, which was used as a hybrid with optimized hyperparameters GPR, responded faster to environmental changes and achieved MPPT with higher efficiency. In addition, in this study, the proposed hybrid method was evaluated in a more complex system using a four series, one parallel PV connection structure. The situations that can be encountered in real time were scenario separately and the performance of the proposed hybrid method was evaluated.

The innovations and contributions of this study are summarized as follows;

- A new hybrid MPPT method, which is a combination of GPR and TOSMC methods with optimized hyperparameters, which has not been presented before in the literature, is proposed.
- Real-time data ranging from temperature – 19.51°C to +31.50°C and irradiance from 0 W/m^2 to 1034.50 W/m^2 were used.
- In the proposed method, simulation studies were performed for both static and dynamic irradiance, temperature and load values, and it was observed that it showed superior performance compared to other algorithms.
- Proposed hybrid method was compared with INC algorithm and GWO algorithm and MPP tracking was achieved with higher efficiency.
- For all different scenarios, the MPPT efficiency value was observed as minimum 99.210 % and maximum 99.913 % with the proposed hyperparameters optimized GPR-HOSMC method.

This paper is organized as follows; Section 2 presents the PV system, Section 3 presents TOSMC, Section 4 presents the optimizable GPR and the proposed method, Section 5 presents simulation results and comparisons, and Section 6 presents conclusions.

2. PV system

PV cells are components that directly generate electrical energy from solar irradiance. Silicon and other semiconductor-related materials are used to build them. When PV cells are exposed to sunlight, their fundamental mechanism of operation is the generation of current. The most commonly used PV cells are monocrystalline, polycrystalline, thin film, bio photovoltaic and bilayer PV cells. Each PV cell model has its own advantages and disadvantages and the user determines the PV cell model by considering factors such as cost, efficiency, production process. The single diode PV cell model, which is the most commonly used PV cell model in the literature, is shown in Fig. 1 [23].

In Fig. 1, I_{ph} represents the current value generated by the photon falling on the photovoltaic cell. The value of I_{ph} varies depending on the reference temperature, instantaneous temperature and irradiance. G and T represent the irradiance and temperature of the PV cell, respectively. R_p represents the resistance due to leakage current and R_s represents the resistance due to cell connections. The output current value generated from the PV panel is given in Equations (1)–(4) [24].

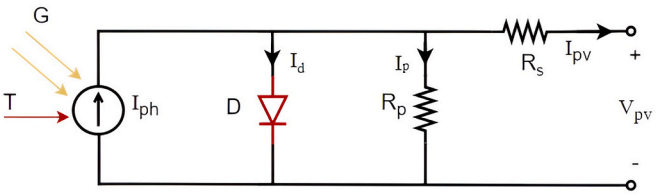


Fig. 1. Single diode cell model.

$$I_{ph} = I_{pv} + I_d + I_p \tag{1}$$

$$I_{ph} = I_o\left[\exp\left(\frac{V_D}{v_t}\right) - 1\right] + \frac{V}{R_p} + I_{pv}\left(\frac{R_s}{R_p} + 1\right) \tag{2}$$

$$I_{pv} = I_{ph}\left(\frac{R_p}{R_p + R_s}\right) - I_o\left(\frac{R_p}{R_p + R_s}\right)\left[\exp\left(\frac{V_D}{v_t}\right) - 1\right] - \frac{V}{R_p + R_s} \tag{3}$$

$$v_t = \frac{k_b A T}{q} \tag{4}$$

In the above equations, I_d represents the diode current, I_p represents the amount of current flowing through R_p in the parallel branch, I_{pv} represents the current value produced by the PV panel, I_o represents the saturation current, V_D represents the diode voltage, k_b represents the Boltzmann constant, A represents the diode ideality factor and q represents the electron charge.

Table 1 shows the specifications of the PV panel (Advance Power API_M250) used in the study.

3. Third order sliding mode controller based MPPT

High-order SMC is used to overcome the steady-state chattering problem, which is one of the most important disadvantages of classical SMC. The aim of high-level SMC is to bring a certain state in the system to the sliding surface. The high level basically includes tracking computation, shifting computation and super twisting computation. The most important advantages of this method over conventional SMC are that it is more robust to uncertainties and external influences, has a faster response time than SMC, and has less fluctuation in the control signal known as chattering. The control signal of TOSMC is defined in Equations (5)–(8).

$$u_1(t) = \lambda_1 \sqrt{|S|}\,sign(S) \tag{5}$$

$$u_2(t) = \lambda_2 \int sign(S)\, dt \tag{6}$$

$$u_3(t) = \lambda_3 sign(S) \tag{7}$$

$$u(t) = \lambda_1 \sqrt{|S|}\,sign(S) + \lambda_2 \int sign(S)dt + \lambda_3 sign(S) \tag{8}$$

Table 1
The technical details of the Advance Power API_M250 Solar module.

Parameter	Specification
Pmax	250.002 W
Voc	37.62 V
Vmp at MPP	30.6 V
Isc	8.59 A
Imp at MPP	8.17 A
Isc (%/deg.C)	0.053725
Voc (%/deg.C)	−0.3564
Cell per module (Ncell)	72

In the above equations, λ_1, λ_2 and λ_3 are fixed parameters used to improve the response of the system and tune the control performance.

3.1. Sliding surface

The sliding surface is selected using Equation (9) [25] to guarantee that the system states reach the sliding surface and to produce the greatest amount of power.

$$\frac{\partial P_{pv}}{\partial V_{pv}} = 0 \tag{9}$$

If the power expression is written in terms of current and voltage as in Equation (10) and substituted in Equation (9), Equation (11) is obtained.

$$P_{pv} = V_{pv} I_{pv} \tag{10}$$

$$\frac{\partial V_{pv} I_{pv}}{\partial V_{pv}} = 0 \tag{11}$$

Rearranging Equation (11) gives Equation (12).

$$\frac{\partial P_{pv}}{\partial V_{pv}} = I_{pv} + V_{pv} \frac{\partial I_{pv}}{\partial V_{pv}} \tag{12}$$

Equation (13) defines the sliding surface using Equation (12).

$$S = \frac{I_{pv}}{V_{pv}} + \frac{\partial I_{pv}}{\partial V_{pv}} \tag{13}$$

3.2. Controller design

For MPPT, the controller for reaching and staying on the sliding surface consists of two terms. These are the super twisting control (u_{TOSMC}) term and the equivalent control term (u_{eq}). The general controller sign is given in Equation (14) [26].

$$u(t) = u_{eq} + u_{TOSMC} \tag{14}$$

Equation 15's equivalent control term can be found by solving the expression [27].

$$\dot{S} = \left[\frac{dS}{d\dot{X}}\right]^T \dot{X} = 0 \tag{15}$$

In the above equation, X represents the state variables of the step-up converter. The state variables in the boost converter are inductance current (I_{pv}) and capacitor voltage (V_0). These state variables are determined by the on and off time of the semiconductor switching element in the boost converter circuit topology. The equation of state for the boost converter circuit topology is shown in Equation (16).

$$\dot{X} = \begin{bmatrix} \frac{dV_0}{dt} \\ \frac{dI_L}{dt} \end{bmatrix} = \begin{bmatrix} -\frac{1}{RC} & \frac{1-D}{C} \\ -\frac{1-D}{L} & 0 \end{bmatrix} \begin{bmatrix} V_0 \\ I_L \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{L} \end{bmatrix} V_{in} \tag{16}$$

In the above equations, D represents the duty cycle of the semiconductor switch, V_0 is the output voltage of the converter topology, V_{in} is the input voltage.

The calculation of the equivalent control term is shown in Equation (17) [21].

$$u_{eq} = -\frac{\left[\frac{dS}{d\dot{X}}\right]^T \begin{bmatrix} \frac{RI_L - V_0}{RC} \\ \frac{V_{in} - V_0}{L} \end{bmatrix}}{\left[\frac{dS}{d\dot{X}}\right]^T \begin{bmatrix} -\frac{I_L}{C} \\ \frac{V_0}{L} \end{bmatrix}} = \frac{V_0 - V_{in}}{V_0} \tag{17}$$

The control term (u_{TOSMC}) is given in the following equation [28].

$$u_{TOSMC} = \lambda_1 \sqrt{|S|} sign(S) + \lambda_2 \int sign(S) dt + \lambda_3 sign(S) \tag{18}$$

The control signal used in the study is given in Equation (19).

$$u(t) = u_{eq} + u_{TOSMC} = \frac{V_0 - V_{in}}{V_0} + \lambda_1 \sqrt{|S|} sign(S) + \lambda_2 \int sign(S) dt + \lambda_3 sign(S) \tag{19}$$

For the convergence property and stability analysis of the TOSMC, the Lyapunov function is determined as in Equation (20) [21].

$$V(t) = \frac{1}{2} S^2(t) \tag{20}$$

The initial and boundary conditions for the Lyapunov function are presented in Equation (21).

$$\left.\begin{array}{c} V(0) = 0 \\ S(t) \neq 0 \text{ için } V(t) > 0 \\ \dot{S}(t) \neq 0 \end{array}\right\} \tag{21}$$

The fact that the derivative of the Lyapunov function is less than zero and the following conditions are met guarantees that the stability state is reached.

$$\left.\begin{array}{l} \dot{V}(t) \langle 0 \\ S(t) \neq 0 \\ \dot{S}(t) \neq 0 \end{array}\right\} \tag{22}$$

When Equality 22 is rearranged, the stability condition can be written as follows.

$$S\dot{S} < 0 \tag{23}$$

The first order derivative of the sliding surface is shown in Equation (24).

$$\dot{S} = \left[\frac{\partial S}{\partial \dot{X}}\right]^T \dot{X} = \left[\frac{\partial S}{\partial I_{pv}}\right] \left(-\frac{V_0}{L}(1-D) + \frac{V_{pv}}{L} \right) \tag{24}$$

The mathematical expression for X is given in Equation (25).

$$X = \begin{bmatrix} I_{pv} & V_0 \end{bmatrix}^T \tag{25}$$

When the sliding surface is written instead, $\left[\frac{\partial S}{\partial I_{pv}}\right]$ is obtained as in Equality 26.

$$\left[\frac{\partial S}{\partial I_{pv}}\right] = \frac{\partial}{\partial I_{pv}} \left[\frac{I_{pv}}{V_{pv}} + \frac{\partial I_{pv}}{\partial V_{pv}}\right] \tag{26}$$

In Equality 26, the expression $\frac{\partial}{\partial I_{pv}} \left[\frac{I_{pv}}{V_{pv}}\right]$ is obtained as in Equality 27.

$$\frac{\partial}{\partial I_{pv}} \left[\frac{I_{pv}}{V_{pv}}\right] = \frac{1}{V_{pv}} - \frac{I_{pv}}{{V_{pv}}^2} \frac{\partial V_{pv}}{\partial I_{pv}} \tag{27}$$

In Equality 26, the expression $\frac{\partial}{\partial I_{pv}} \left[\frac{\partial I_{pv}}{\partial V_{pv}}\right]$ is expressed as in Equality 28.

$$\frac{\partial}{\partial I_{pv}} \left[\frac{\partial I_{pv}}{\partial V_{pv}}\right] = \frac{q}{k_b T A} \frac{\partial I_{pv}}{\partial V_{pv}} \frac{\partial V_{pv}}{\partial I_{pv}} \tag{28}$$

When all equations are written in place, the expression $\left[\frac{\partial S}{\partial I_{pv}}\right]$ is obtained as in Equality 29.

$$\left[\frac{\partial S}{\partial I_{pv}}\right] = \frac{1}{V_{pv}} - \frac{I_{pv}}{{V_{pv}}^2} \frac{\partial V_{pv}}{\partial I_{pv}} + \frac{q}{k_b T A} \frac{\partial I_{pv}}{\partial V_{pv}} \frac{\partial V_{pv}}{\partial I_{pv}} \tag{29}$$

The derivative of the PV current is expressed in Equation (30).

$$\frac{\partial I_{pv}}{\partial V_{pv}} = -\frac{q}{k_b TA} I_d \exp\left(\frac{q}{k_b TA}\right) < 0 \tag{30}$$

PV voltage is given in Equation (31).

$$V_{pv} = \frac{k_b TA}{q} \ln\left(\frac{I_{ph} + I_d - I_{pv}}{I_d}\right) \tag{31}$$

The first derivative of the PV voltage depending on the current is given in Equation (32).

$$\frac{\partial V_{pv}}{\partial I_{pv}} = -\frac{k_b TA}{q} \frac{I_d}{I_{ph} + I_d - I_{pv}} < 0 \tag{32}$$

The sign of the first term is determined using Equation (33).

$$\frac{\partial S}{\partial I_{pv}} > 0 \tag{33}$$

The sign of the expression $\left(-\frac{V_0}{L}(1-D)+\frac{V_{pv}}{L}\right)$ is verified using the following equations.

$$\left.\begin{aligned} \dot{X} &= \left(-\frac{V_0}{L}(1-D)+\frac{V_{pv}}{L}\right) \\ \dot{X} &= \left(-\frac{V_0}{L}(1-(u_{eq}+u_{TOSMC}))+\frac{V_{pv}}{L}\right) \\ \dot{X} &= \left(-\frac{V_0}{L}\left(1-\left(\frac{V_0-V_{in}}{V_0}+u_{TOSMC}\right)\right)+\frac{V_{pv}}{L}\right) \\ \dot{X} &= \left(\frac{V_0}{L}(u_{TOSMC})\right) \\ \dot{X} &= \left(\frac{V_0}{L}\left(\lambda_1\sqrt{|S|}\,sign(S)+\lambda_2\int sign(S)dt+\lambda_3 sign(S)\right)\right) \end{aligned}\right\} \tag{34}$$

When the equations are rearranged, the stability expression is obtained as in Equation (35).

$$S\dot{S} = S\left[\frac{\partial S}{\partial I_{pv}}\right]\left(\frac{V_0}{L}\left(\lambda_1\sqrt{|S|}\,sign(S)+\lambda_2\int sign(S)dt+\lambda_3 sign(S)\right)\right) \tag{35}$$

The stability condition according to the expression $S\dot{S} < 0$ is given in Equation (36).

$$\lambda_1\sqrt{|S|}sign(S)+\lambda_2\int sign(S)dt+\lambda_3 sign(S) < 0 \tag{36}$$

According to Equation (36), the system is stable since the values of λ_1, λ_2 and λ_3 are negative for stability.

4. Gaussian process regression (GPR)

GPR is a frequently preferred regression method in statistics and machine learning. This method is particularly suitable for systems with small data sets or data sets with complex non-linear relationships. The most important advantages of this method are flexibility, efficiency with less data, tunability of hyperparameters and the ability to model non-linear relationships more successfully than other methods. The GPR model based on Gaussian distribution for the function space $f(x) = x^T w$ and the data set to be trained is $[f(x_1), f(x_2), f(x_3), \ldots, f(x_n)]$ is given in Equation (37) [29].

$$f(x) \sim GP(m(x), k(x, x')) \tag{37}$$

In the above equation, x represents the training data set, $m(x)$ represents the mean value of the training data set and $k(x, x')$ represents the covariance function. The mathematical expressions for $m(x)$ and $k(x, x')$ are shown in Equation (38).

$$\left.\begin{aligned} m(x) &= E[f(x)] \\ k(x,x') &= E[(f(x)-m(x))(f(x')-m(x'))] \end{aligned}\right\} \tag{38}$$

The output expression obtained with the GPR model is given below [29];

$$y = x^T w + \varepsilon, \varepsilon \sim N(0, \sigma^2) \tag{39}$$

In the above equation, y is the output vector of the GPR, ε is the noise expression, and w is the coefficient associated with the x data. The covariance function is usually expressed in the literature as in Equation (40) [29].

$$k(x, x') = He^{\left(\frac{x-x'}{2\lambda}\right)} \tag{40}$$

In Equation (40), λ and H are hyperparameters. Hyperparameters are known as parameters that cannot be learned from the data but affect the behavior of the model. λ is the parameter that determines the attenuation rate value. This parameter shows how a value decreases over time. H is used to determine and control how strong the relationship between two variables will be. Matlab/Regression Toolbox was used to optimize the hyperparameters in the study.

4.1. Training results

The suggested hybrid method's performance was assessed using real-time data. This real-time data was obtained from NASA (Latitude:39.901, Longitude:41.2501). The data covering a time period of three months was trained using GPR with optimized hyperparameters. The performance metrics that were used to demonstrate the effectiveness of the trained model were Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Mean Squared Error (MSE) and Coefficient of determination (R^2). The mathematical expressions for these criteria are given in Equations (41)–(44).

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(y_i - \hat{y}\right)^2} \tag{41}$$

$$MSE = \frac{1}{N}\sum_{i=1}^{N}\left(y_i - \hat{y}\right)^2 \tag{42}$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N}\left|y_i - \hat{y}\right| \tag{43}$$

$$R^2 = 1 - \frac{\sum\left(y_i - \hat{y}\right)^2}{\sum\left(y_i - \bar{y}\right)^2} \tag{44}$$

In the above equations, y_i represents the actual data, $\hat{y}$ the predicted values and $\bar{y}$ the average value of the actual data. During model training, irradiance and temperature values were set as input and voltage value as output. The data was standardized using Equation (45) due to the differences between the variables while training.

$$Xstand = \frac{x - mean(x)}{standart\ deviation(\sigma)} \tag{45}$$

Table 2 displays the outcomes of the training phase.

The standardized actual data and the values predicted by the model are shown in Fig. 2.

The actual and estimated values for standardized data and the perfect prediction values of these values are shown in Fig. 3.

The difference between standardized data and predicted values is shown in Fig. 4.

Table 2
Training results obtained using GPR with optimized hyperparameters.

Results	RMSE	MSE	MAE	R^2
Hyperparameters optimized GPR	0.082239	0.0067633	0.030472	0.99

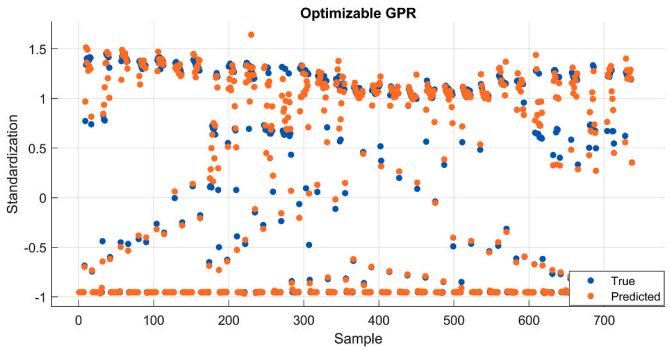


Fig. 2. Standardized real data and model predicted values.

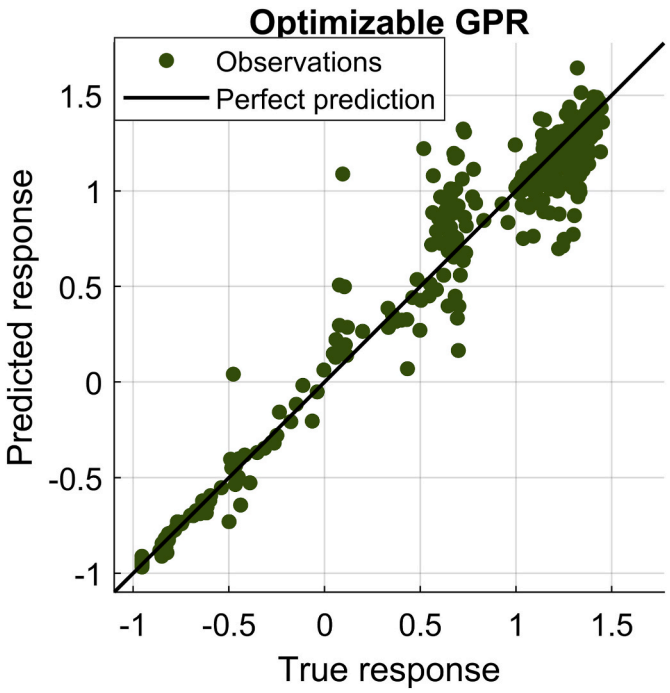


Fig. 3. Best prediction values.

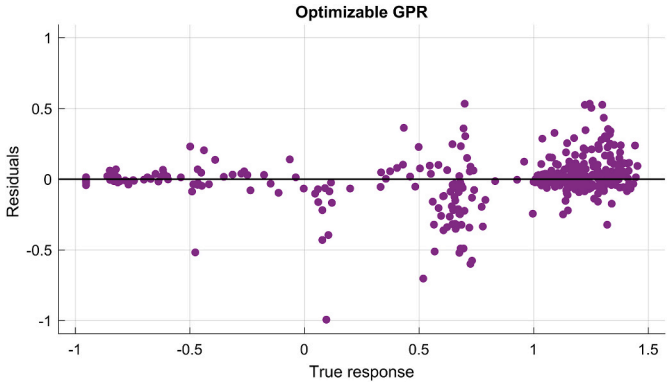


Fig. 4. Difference between observed values and predicted values.

5. Simulation results and comparisons

Simulation studies were carried out using the Matlab/Simulink program for the same sampling time (0.00001 s). Advance Power API_M250 Solar module was used as a PV panel in the study. The electrical properties of this panel are given in Table 1. In addition, a four-series parallel (4Sx1P) PV connection structure was preferred for the performance of the proposed hybrid method in the study. The Simulink block diagram of the designed system to demonstrate the effectiveness of the proposed hybrid method is shown in Fig. 5. During the simulation studies, four PV panels were used to examine the effect of partial shading. In order to increase the voltage value obtained from PV panels to the voltage value desired by the load, the boost converter topology whose values are given in Table 3 was used. Optimizable GPR and TOSMC are combined to control the duty cycle of the semiconductor switching element, allowing the system to operate at the GMPP. To illustrate the efficacy of the suggested hybrid approach, six different scenarios involving high, medium, and low temperatures and irradiance levels are examined. Fig. 6 displays the P-V characteristics of the PV panels under various scenarios.

The performance of the proposed hybrid method is compared with GWO and INC algorithms in six different scenarios. Table 4 shows the temperature and irradiance data for the first and third scenarios.

5.1. First scenario: high level of irradiance and temperature values ($G > 850$ W/m2, $T > 25$ °C)

As the first scenario, in order to demonstrate the effectiveness of the proposed method, the temperature and irradiance values of August, a month with high irradiance and temperature values, were preferred. The temperature and irradiance values used for this scenario are given in Table 4. As shown in Fig. 6, the local maximum power point for this scenario is calculated as 681 W and the global maximum power point is calculated as 884.57 W. For this scenario with partial shading, the proposed hybrid method reached the maximum power point (882.13 W) in 0.183s. While the INC algorithm followed the value of the local maximum power point and reached 658 W, the GWO algorithm reached 857.55 W. The power dissipated on the load for the proposed method, GWO, and INC algorithms is displayed in Fig. 7-a. Fig. 7-b and Fig. 7-c display the voltage and current values of the load for the suggested hybrid method (210.02V-4.2 A), GWO (207.14V-4.14 A), and INC (181.37V-3.62 A), respectively.

5.2. Second scenario: medium level of irradiance and low level of temperature (550 W/m2 $< G < 710$ W/m2, $T < 3$ °C)

As the second scenario, data from February, when irradiance values are at medium level and temperature values are at low level, were used. The temperature and irradiance values used for this scenario are given in Table 4. In the studies on MPPT in the literature, it is generally observed that the irradiance values change while the temperature value is constant. The cold winter weather in the area where the data was collected was considered when creating the scenarios. In Fig. 6, the local maxima are 181.6 W, 362 W and 522 W and the global maximum power point is 670.76 W. The power, voltage and current values for the proposed hybrid method (669.71 W, 182.99 V, 3.659 A, 0.193s), GWO algorithm (657.17 W, 181.27 V, 3.625 A) and INC algorithm (170.23 W, 92.26 V, 1.845 A) are shown in Fig. 8-a, b, c. In the second case, the proposed approach and the GWO algorithm arrived at the global power point, while the INC algorithm followed the local maximum power point. Compared to the GWO algorithm, the hybrid method that has been suggested is more efficient and tracks GMPP more quickly.

5.3. Third scenario: low level of irradiance and medium level of temperature ($G < 470$ W/m2, 9 °C $< T < 13$ °C)

As the last scenario, the temperature and irradiance values for October, when the irradiance levels are low and the temperature levels are medium, were used. The irradiance and temperature values determined for this scenario are given in Table 4. For the case of the irradiance and temperature values determined in Table 4, the local maximum

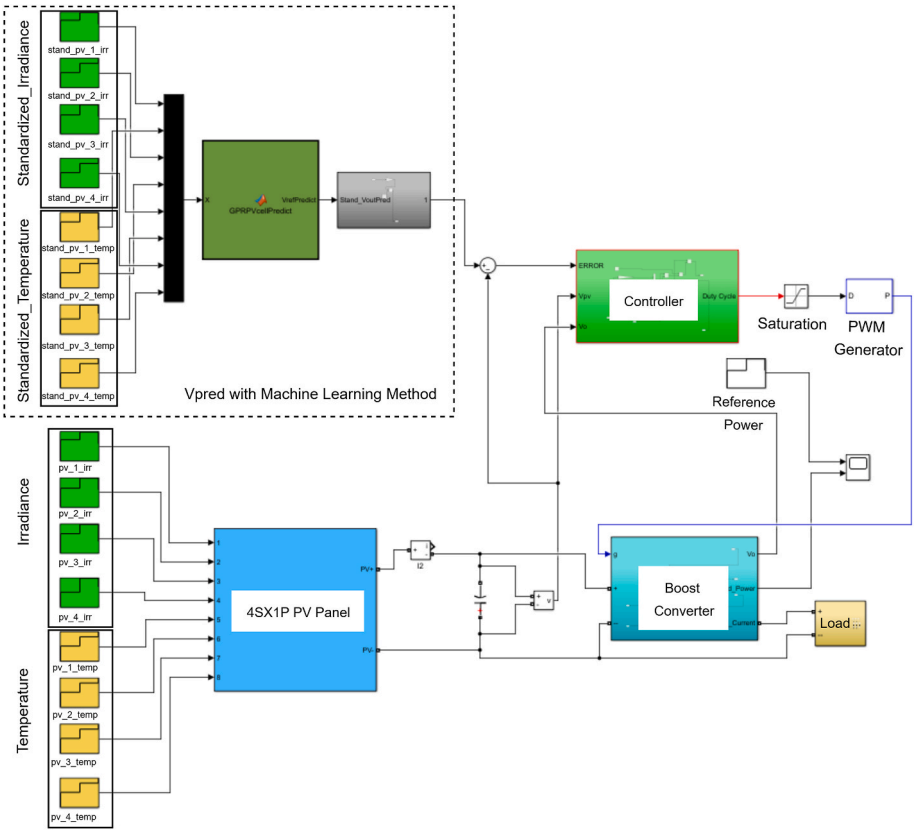


Fig. 5. Overall block diagram of the proposed system in Matlab/Simulink.

Table 3
The electrical component values in the topology of the boost converter.

Parameter	Value
C_{in} and C	0.1 μF- 1.659 mF
L	0.0284 *H*
f	50 kHz
R	50 Ω

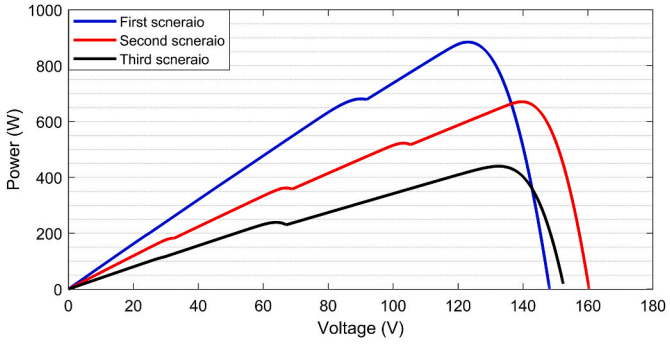


Fig. 6. Power-voltage characteristics for different scenarios.

power point is calculated as 238.9 W and the global maximum power point is calculated as 439.87 W. The proposed hybrid method achieved 439.376 W at 99.887 % efficiency, 0.216s. The GWO algorithm achieved 433.702 W at 98.597 % efficiency. The classical optimization algorithm INC reached 230.05 W local maximum power point with 52.299 % efficiency. Fig. 9-a shows the power-time variation of the proposed hybrid method and algorithms. Fig. 9-b and Fig. 9-c shows the changes in

Table 4
Temperature and irradiance values for various scenarios and MPP.

Scenarios	PV Panel	Irradiance (W/m^2)	Temperature (°C)	MPP (W)
1st scenario	PV-1	939.47	29.30	884.57
	PV-2	925.98	28.72	
	PV-3	919.88	27.50	
	PV-4	853.17	27.12	
2nd scenario	PV-1	574.44	0.32	670.76
	PV-2	701.10	1.29	
	PV-3	608.16	1.64	
	PV-4	654.30	2.88	
3rd scenario	PV-1	467.66	9.72	439.87
	PV-2	453.66	12.17	
	PV-3	399.29	11.82	
	PV-4	399.29	11.82	

voltage and current on the load.

The boost converter output voltage and current values in the simulation studies performed for three different scenarios are given in Table 5.

The results of all scenarios are given in Table 6 comparatively.

The efficiency values of the proposed hybrid method, GWO and INC algorithms under different partial shading conditions are shown in Fig. 10.

5.4. Fourth scenario: different location of GMPP in P-V characteristic

In the fourth scenario, the performance of the proposed method was evaluated for the case where the GMPP value is in the middle of the P-V characteristic, unlike the other three scenarios. For the other three scenarios, the GMPP value is located at the rightmost part of the P-V characteristic.

Fig. 11-a presents the P-V characteristics of the PV system for the

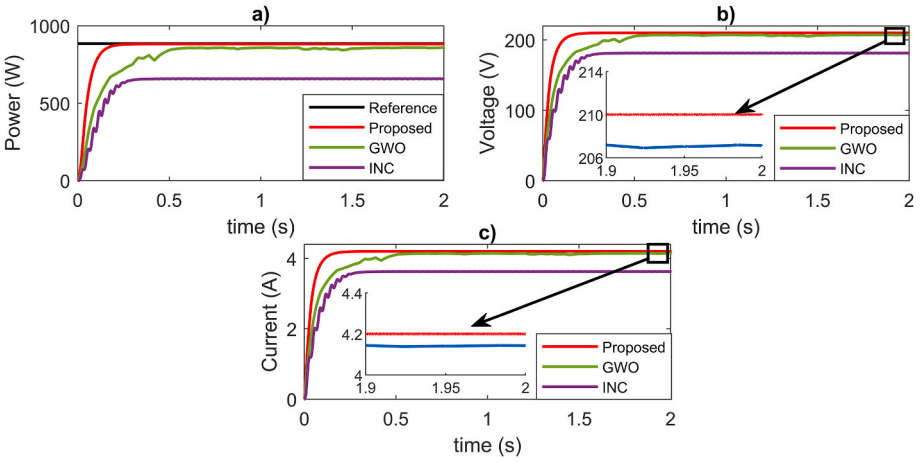


Fig. 7. Simulation results for the first scenario; a-load power, b-load voltage, c-load current.

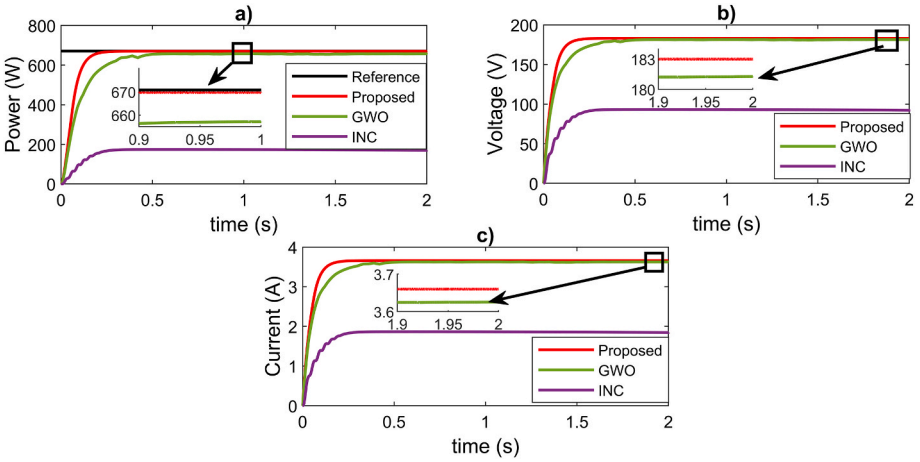


Fig. 8. Simulation results for the second scenario; a-load power, b-load voltage, c-load current.

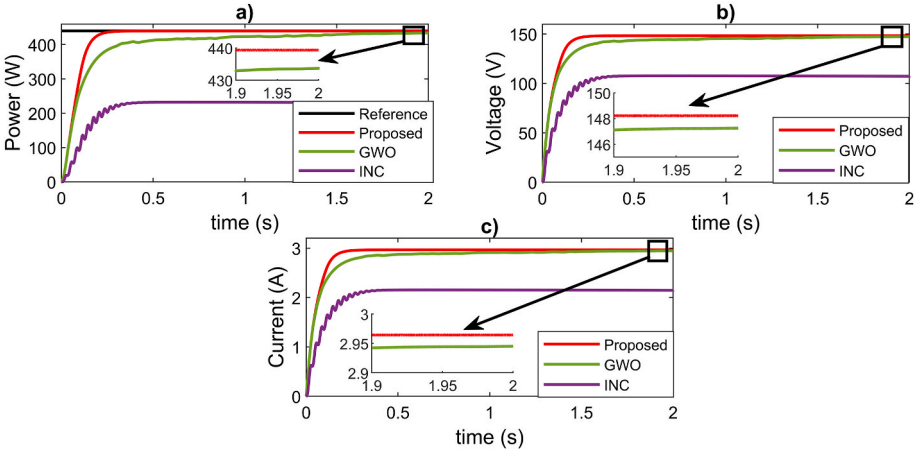


Fig. 9. Simulation results for the third scenario; a-load power, b-load voltage, c-load current.

irradiance values in Table 7. Here, it is seen that the power value in the MPP is 633.27 W and the voltage value is 94.7647 V. The fact that this voltage value in the MPP is at lower levels than the other scenarios were considered when creating the scenario. The changes in the power, voltage and current of the load with respect to time for the application of the proposed method are shown in Fig. 11-b, Fig. 11-c and Fig. 11-d, respectively. The results of this scenario are given in detail in Table 7.

5.5. *Fifth scenario: dynamic change of irradiance and temperature*

In the fifth scenario, the situation where the irradiance and temperature values of the panels change dynamically at 2s intervals was evaluated. Table 7 shows the temperature and irradiance values of the panels for each time period. The results of this scenario are shown in Fig. 12.

Table 5
The real-time data-based simulation results for various partial shading conditions.

Scenarios	Methods	Load Voltage (V)	Load Current (A)
1st scenario	Proposed	210.02	4.2
	GWO	207.14	4.14
	INC	181.37	3.62
2nd scenario	Proposed	182.99	3.659
	GWO	181.27	3.625
	INC	92.26	1.845
3rd scenario	Proposed	148.22	2.964
	GWO	147.25	2.945
	INC	107.25	2.144

Table 6
The real-time data-based simulation results for various partial shading conditions.

Scenarios	Methods	GMPP(W)	Measured Power (W)	Steady State Error (W)	Efficiency (%)
1st scenario	Proposed	884.57	882.13	2.44	99.724
	GWO	884.57	857.55	27.02	96.945
	INC	884.57	658	226.57	74.386
2nd scenario	Proposed	670.76	669.71	1.05	99.843
	GWO	670.76	657.17	13.59	97.973
	INC	670.76	170.23	500.53	25.378
3rd scenario	Proposed	439.87	439.376	0.494	99.887
	GWO	439.87	433.702	6.168	98.597
	INC	439.87	230.05	209.82	52.299

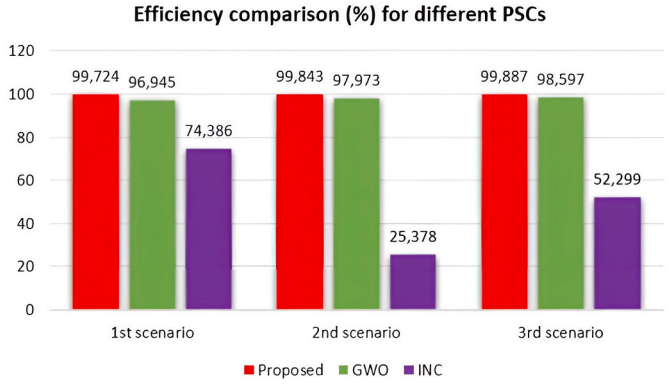


Fig. 10. The efficiency values of the proposed hybrid method, GWO and INC algorithms.

It has been observed that the proposed method performs MPPT with high efficiency for this scenario. The results of this scenario are presented in detail in Table 7.

5.6. Sixth scenario: dynamic change of irradiance-temperature and load

In the sixth scenario, the performance of the proposed method was evaluated when both the irradiance and temperature values and the load changed. For this scenario, the irradiance and temperature values were changed in 1 s, and the load was changed in 0.5 s. The results of this scenario are shown in Fig. 13. For all different cases, the proposed method performed MPPT quickly with high efficiency. Detailed results of this scenario are given in Table 8.

The comparison between the suggested hybrid method and the literature studies is displayed in Table 9. The fact that the DC-DC converter topology utilized between the PV panels and the load is the same was considered during the comparison. Furthermore, performance metrics like convergence time, conditions, and efficiency are used to base the comparison.

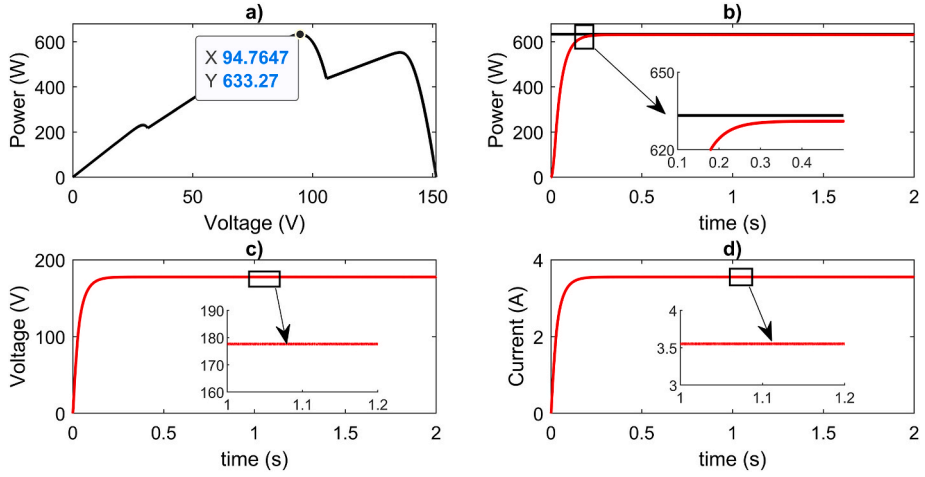


Fig. 11. Simulation results for the fourth scenario; a-P-V characteristic, b-load power-time, c-load voltage-time, d-load current-time.

Table 7
Power and efficiency obtained for irradiance and temperature values in the fourth and fifth scenarios.

Scenarios	Irradiance (W/m^2)/Temperature (°C)	Time (s)	GMPP (W)	Measured Power (W)	Efficiency (%)
4th scenario	(PV-1) 478.97/17.16 (PV-2) 972.80/20.55 (PV-3) 808.01/20.56 (PV-4) 808.01/20.56	0–2	633.27	631.06	**99.651**
5th scenario	(PV-1) 690.63/5.46 (PV-2) 617.02/11.92 (PV-3) 454.39/11.51 (PV-4) 632.59/4.69	0–2	429.06	428.69	**99.913**
	(PV-1) 858.58/7.79 (PV-2) 772.84/14.17 (PV-3) 574.50/12.64 (PV-4) 472.59/6.37	2–4	555.38	554.71	**99.879**
	(PV-1) 969.08/10.07 (PV-2) 855.70/15.70 (PV-3) 691.41/13.47 (PV-4) 605.30/8.23	4–6	699.78	698.64	**99.837**
	(PV-1) 1025/12.08 (PV-2) 936.97/16.76 (PV-3) 728.17/13.64 (PV-4) 609.48/9.48	6–8	706.86	705.65	**99.828**
	(PV-1) 1035/13.56 (PV-2) 991.31/17.37 (PV-3) 691.73/12.69 (PV-4) 595.56/9.98	8–10	690.55	689.43	**99.837**
	(PV-1) 990.43/14.60 (PV-2) 833.73/17.70 (PV-3) 620.70/11.16 (PV-4) 522.01/9.81	10–12	606.68	605.84	**99.861**
	(PV-1) 889.72/15.07 (PV-2) 663.87/17.59 (PV-3) 158.30/9.59 (PV-4) 357.89/9.18	12–14	357.66	356.20	**99.591**

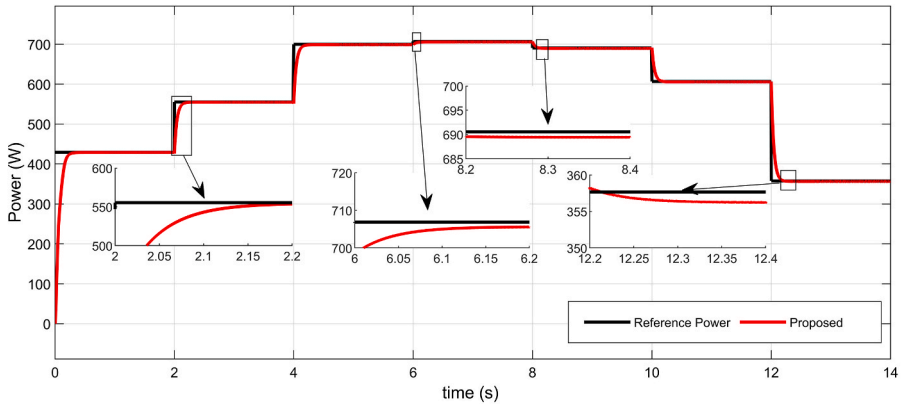


Fig. 12. Output power for dynamic irradiance and temperature values.

6. Conclusions

This study used a hybrid approach that can perform MPPT quickly and accurately in PV systems under partial shading conditions is presented. A hybrid method consisting of hyperparameters optimized GPR and TOSMC is proposed in this study. In the suggested hybrid method, the hyperparameters optimized GPR method is used to determine the closest operating point to the GMPP, and then MPPT is performed using the TOSMC. This method performs MPPT quickly and effectively under partial shading conditions, while requiring less computation compared to metaheuristic optimization algorithms. This study compares the suggested algorithm for six scenarios with GWO and INC. the first scenario, data from August, when the irradiance and temperature values are at the highest values, were used. The recommended approach outperformed the other algorithms in terms of convergence time, achieving an efficiency value of 99.724 % in this particular scenario. Alternatively, INC was unable to surpass the local maximum point and thus not reach the global maximum value. The second scenario uses data from February. In this scenario, the performance of the proposed hybrid method is evaluated in case of low temperature values. In this scenario, the proposed hybrid method achieved *99.843 %* efficiency in *0.193* s. The three scenario uses data from October. While determining this scenario, the average values of irradiance and temperature were taken into consideration. In this scenario, the proposed hybrid method achieved *439.376W* in *0.216 s* with *99.887 %* efficiency. GWO reached *433.702W* in *1.581 s* with *98.597 %* efficiency and INC reached *230.05W* at the local power point. In addition, in order to analyse the performance of the proposed method under different partial shading conditions, scenarios were created for dynamic irradiance-temperature, dynamic load status and GMPP occurring at different points. For all scenarios, the

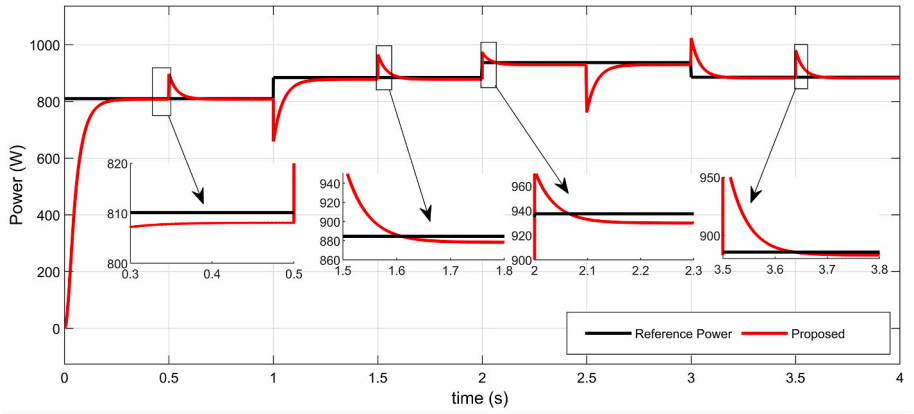


Fig. 13. Output power for dynamic irradiance-temperature and load values.

Table 8
Power and efficiency obtained for load values and irradiance-temperature values in the sixth scenario.

Scenarios	Irradiance (W/m^2)/Temperature (°C)	Time (s)	Load (Ω)	GMPP (W)	Measured Power (W)	Efficiency (%)
6th scenario	(PV-1) 852.96/27.15	0–0.5	50	810.13	808.12	**99.751**
	(PV-2) 851.30/27.03					
	(PV-3) 852.50/25.73	0.5–1	40			
	(PV-4) 769.59/25.75					
	(PV-1) 939.47/29.30	1–1.5	60	884.57	878.3	**99.291**
	(PV-2) 925.98/28.72					
	(PV-3) 919.88/27.50	1.5–2	50			
	(PV-4) 853.17/27.12					
	(PV-1) 970.41/30.68	2–2.5	40	937.40	930	**99.210**
	(PV-2) 968.80/29.78					
	(PV-3) 951.48/28.65	2.5–3	60			
	(PV-4) 934.65/28.05					
	(PV-1) 921.89/31.36	3–3.5	50	885.62	883.1	**99.715**
	(PV-2) 919.85/30.29					
	(PV-3) 892.07/29.26	3.5–4	40			
	(PV-4) 886.85/28.48					

Table 9
Comparison of the performance of the proposed method.

Algorithm	DC-DC Converter	Tracking Time (s)	PV Panel Conditions	Efficiency (%)
Method in GWO [5]	Boost Converter	–	Partial Shading	99.81
Method in hybrid [30]	Boost Converter	0.13	Partial Shading	99.55
Method in TOSMC [22]	Boost Converter	0.02	Variable Climatic Conditions	93
Method in MVP [31]	Boost Converter	0.447	Partial Shading	98.82
Method in SSO [32]	Boost Converter	0.2	Partial Shading	99.3
Method in MGWO [33]	Boost Converter	0.96	Partial Shading	99.53
Method in APSO [34]	Boost Converter	3	Partial Shading	99.5
Method in EWOA [35]	Boost Converter	0.2	Partial Shading	99.7
Method in SSA [36]	Boost Converter	0.66	Boost Converter	99.48
Proposed Method	Boost Converter	0.183	Partial Shading	99.913

proposed hyperparameters optimized GPR based third order sliding mode controller achieves shorter convergence time and higher efficiency than INC and GWO. With the suggested hybrid method, the disadvantage of classical optimization algorithms, which is tracking the local power point under partial shading conditions, is eliminated. In addition, the suggested hybrid method overcomes the disadvantages of metaheuristic optimization algorithms such as complex mathematical calculations and long convergence time.

CRediT authorship contribution statement

Mehmet Yilmaz: Writing – original draft, Software, Methodology, Conceptualization. **Muhammed Fatih Çorapsiz:** Writing – review & editing, Supervision, Methodology, Conceptualization, Software.

Data availability

Data will be made available on request.

Real time temperature and irradiance data is received from https://power.larc.nasa.gov/data-access-viewer/.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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